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Project Proposal
Industry Based Software Engineering Project (IT2992)

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1. Introduction

Global trends in transportation management are increasingly leaning towards digital solutions in today's expeditious and competitive landscape. As a consequence, online bus management systems have made significant revolution by transforming the way public and private bus services are utilized into a new dimension. Especially for a university setting like Southern Campus located in a rural province, the need for an organized and efficient transportation service is very important. The paucity of transport options in a situation like this further emphasizes the imperative of implementing a robust bus management system to ensure reliable and regular services for students and staff. As a facility provided for this need, currently the Southern Campus of Kotelawala Defence University is conducting a transport service that provides transportation to students and lecturers from the university premises to Kadawatha on weekends.

It is evident that, although the existing manual transport system service fulfills the user's transport needs, it still faces several challenges that affect its overall effectiveness. Among the major difficulties faced by the current transport service are primarily, difficulties in maintaining effective communication, the lack of real-time updates and the inability to do so even in situations that could easily be facilitated due to malfunctions in the administrative system could basically seen. Consequently, this project intends to revive the system based primarily on overcoming the obstacles of the staff bus management system and develop a comprehensive Staff Bus Management System that addresses these issues by leveraging modern technology. The proposed system will include features such as real-time GPS tracking, an online seat booking and returning platform, automated notifications, and feedback mechanisms to enhance user experience and operational efficiency.

By implementing this system, the main aim is to provide a more reliable and user-friendly transportation management solution, improving the weekly commute for students and staff. This proposal outlines the project’s objectives, methodology, timeline, and expected outcomes, providing a detailed plan for the successful execution of the Staff Bus Management System.

1.1. Client Details

What is KDUSC Staff Bus?

The KDUSC Staff Bus is the transportation method which is maintained by Southern Campus of Kotalawala Defence University and Controlled under the supervision of g/579/63 Sergeant Gunasekara KMSP. They use a bus taken from GPL Priyankara Transport Service on payment basis for transportation. The service has been in usage for over three years, having commenced in June 2021.From that day until now, they have used 3 buses which were obtained on a payment basis and currently the 3rd bus is being used in the Staff Bus service. During the period from then to today, two drivers were involved with this service and currently the second driver is engaged in the service. At first, they provided this service only for lecturers and for almost a year they extended this service to day scholars as well.

The weekend transport service of Southern Campus of Kotelawala Defence University uses the expressway as the route to traverse from the university premises to Kadawatha. Here, the administration of KUSC Staff bus will charge one thousand rupees for all the exits from Southern Campus to Galle, and two thousand rupees for each exit from there to Kadawata. Generally, a ticket is issued for each journey by the bus management located in the headquarters building on Southern Campus. According to the current situation, all the relevant records in the required files are maintained manually, which consumes a lot of time and stationery.

2. Background and Motivation

The KDUSC Staff Bus transport service has now become an essential service for students and lecturers who use the staff bus as weekend transportation. This service regulates according to a fixed schedule, with the bus departing from the Southern Campus on Saturdays and travel along to Kadawatha after that the return journey commences on Tuesday mornings from Kadawatha, transporting passengers (Students and Staff) back to the Southern Campus. Despite its importance, the existing system lacks real-time updates, making it difficult for users to get accurate information about bus schedules and availability. Bus service information such as timetables and pick up or drop off spots are mainly accessed through WhatsApp groups and word of mouth or by calling the driver, which often leads to communication gaps and uncertainty among users. As this project have understood mainly these limitations highlight the need for a more efficient and reliable transport management solution.

The motivation behind creating new software for managing The Staff Bus is centered on a user-focused approach that seeks to meet the needs and expectations of both students and lecturers, thereby enhancing their daily commuting experience. This user-friendly system will improve operational efficiency by optimizing resource utilization, reducing operational costs, and refining scheduling, seat allocation, and route management through informed decision-making. Safety and convenience will be prioritized with features such as real-time tracking, emergency alerts, online seat reservations and automated notifications, making the service more reliable and secure. Furthermore, it aligns with sustainability objectives by minimizing paper consumption through digital ticketing. This project also offers educational and developmental benefits for the project team, providing hands-on experience with real-world problem-solving, project management, and the application of IT skills.

3. Problem Domain

The staff bus service at Kotelawala Defence University (KDU) Southern Campus faces several challenges that significantly impact its users, including students and lecturers who depend on it for their weekly commutes. One of the primary issues is the absence of real-time updates, which creates uncertainty regarding bus schedules and departure times. Many students have reported missing buses due to outdated or inaccurate information. Furthermore, communication gaps between the administration and users often result in confusion, particularly when there are changes to the schedule. This problem is worsened by the lack of a centralized platform for obtaining bus-related information. Additionally, overcrowding and insufficient seating are common, leading to discomfort and safety concerns during travel, especially when passengers board without prior seat reservations. Surveys and interviews with regular bus users indicated that these issues lead to significant inconvenience, missed appointments, and general dissatisfaction with the service. The data revealed that 39.4% of respondents have experienced problems with the current staff bus service, while 78.8% indicated a desire for an online booking system to simplify the process. Observational studies further supported these findings, underscoring the inefficiencies and logistical difficulties of the current system. Addressing these concerns is essential for improving the reliability, safety, and overall user experience of the KDU staff bus service.

Chart 0-1: Survey Result

Chart 0-2: Survey Result

4. Aim

The main objective of this project is to create an inclusive Staff Bus Management System that resolves the existing challenges encountered by students and lecturers at Kotelawala Defense University, Southern Campus, thereby enhancing their weekly commuting experience. This system seeks to deliver reliable and user-friendly transportation services by integrating real-time GPS tracking, online seat reservations, automated alerts, and feedback options. By achieving these goals, this project will significantly improve operational efficiency, safety, and convenience for users, aligning with KDU's overarching mission of utilizing innovative solutions to enrich campus life. By tackling communication barriers, the absence of real-time updates, and administrative inefficiencies, the proposed system will provide a streamlined and effective transportation service for the university community.

5. Objectives

This project aims to develop an Inclusive Rental Staff Bus Management System for Kotelawala Defence University (KDU)Southern, and it has highlighted some objectives in order to solve the identified problems and enhance user experience, Goals that are intended to provide concrete, actionable steps towards an efficient, reliable transportation service.

1. Develop a Real-Time Bus Tracking System:
    i. Implement GPS tracking to provide real-time location updates of the Staff bus via a mobile app.
    ii. Ensure the system is accurate and updates in real-time to give users reliable information.

2. Create an Online Booking Platform:
    i. Design and develop a user-friendly platform for students and lecturers to book seats according to a seat number, view schedules, and manage their reservations online.

3. Integrate Automated Notifications:
    i. Set up automated notification systems to inform users about bus schedules, delays, cancellations, and other important updates through SMS and push notifications.

4. Implement Ticket Rollover and Reissue System:
    i. Develop a system that allows users to return and reissue unused tickets for future use.
    ii. Ensure that the ticket rollover process is seamless and user-friendly, providing flexibility for students and lecturers.

5. Conduct User Testing and Training:
    i. Perform extensive user testing to gather feedback and ensure the system meets user needs.
    ii. Provide training sessions for students and lecturers to effectively use the new system.

By addressing these objectives, the project will deliver a robust and efficient Staff Bus Management System that meets the needs of the KDU community, ensuring a reliable and user-friendly transportation service.

6. Hypothesis

In the creation of the KDUSC Staff Bus Management System, this propose that the incorporation of real-time GPS tracking, online booking systems, and improved communication channels will greatly enhance operational efficiency and the user experience. The group members of this project believe that real-time tracking will lead to reduce waiting times and better punctuality, while the online booking system will facilitate seat reservations and lessen administrative responsibilities. Furthermore, enhanced communication channels are anticipated to boost user engagement and minimize uncertainty regarding bus schedules. On the other hand, the null hypothesis suggests that these technological advancements will not result in any meaningful improvements in operational efficiency or user satisfaction, indicating that users will have no significant effect on wait times, punctuality, seat reservation procedures, or user engagement. Testing this null hypothesis will be essential in evaluating the effectiveness of the proposed system in achieving the intended outcomes.

7. Literature Review

Research on online transport management systems underscores the substantial influence of real-time tracking, online booking platforms, and enhanced communication channels on both operational efficiency and user satisfaction.

7.1. Real-Time Tracking

The incorporation of real-time tracking technology has resulted in notable enhancements in the efficiency and dependability of transportation systems. Smith et al. (2018) illustrate that GPS-based tracking systems greatly improve university bus services by offering real-time updates on bus locations, which in turn boosts punctuality and user satisfaction. This technology empowers users to track bus movements live, thereby alleviating uncertainty and minimizing waiting times. Lee and Kim (2019) corroborate these findings, highlighting that real-time tracking fosters greater user convenience and trust in transportation services. They also point out challenges, such as the necessity for seamless integration with existing infrastructure, which need to be resolved to fully capitalize on the advantages of real-time tracking. The Online Bus Ticket Reservation System (OBTRS) for Sri Lanka's National Transport Commission (NTC) exemplifies significant advancements by incorporating real-time GPS data, enabling users to track bus locations and schedules, thereby improving both user experience and operational efficiency (Oloyede et al., 2014).

7.2. Online Booking Platforms

Online booking platforms have revolutionized the management of transportation logistics and elevated user experience. Johnson (2020) explores how digital booking systems optimize seat reservations and decrease administrative burdens, emphasizing the effectiveness of online platforms in overseeing seat allocations and bookings. However, Johnson also highlights that technical challenges and a lack of user training can limit the effectiveness of these systems. The OBTRS is a prime example of a well-executed online booking platform, employing technologies such as XHTML, PHP, SQL, Ajax, CSS, and JavaScript to streamline seat management, reservations, and payments. This system reduces manual operations and minimizes errors, catering to the requirements of the KDUSC Staff Bus Management System by illustrating the advantages of a strong online booking solution (Oloyede et al., 2014).

7.3. Enhanced Communication Channels

Effective communication is vital for improving user experience and ensuring prompt updates. Brown et al. (2021) emphasizes the significance of automated notification systems in transport management platforms. Their study shows that automated alerts regarding schedules, delays, and cancellations enhance user experience by delivering timely and precise information. The OBTRS exemplifies this strategy by employing email and SMS to dispatch confirmation messages, ensuring users receive updates via their preferred communication methods. This dual-channel approach boosts the dependability and convenience of the booking process, as evidenced by the OBTRS (Oloyede et al., 2014).

The literature reviewed emphasizes the key elements of efficient transport management systems: real-time tracking, online booking platforms and improved communication channels. These components are essential for creating a system that addresses contemporary transportation needs. Insights garnered from existing systems, like the OBTRS, offer important guidance for integrating these features into the KDUSC Staff Bus Management System. By tackling challenges and adopting best practices identified in earlier studies, the KDUSC system is designed to improve operational efficiency, user satisfaction, and overall service reliability. The goal of the eBus Services system's integration of these features and technologies is to offer Kotelawala Defence University's Southern Campus a dependable, effective, and user-friendly transportation option that will greatly increase campus mobility and user happiness.

8. Methodology

The Methodology section describes the structured approach this project will adopt to fulfill the goals of the Staff Bus Management System project. This project will employ a mixed-methods strategy that integrates both qualitative and quantitative data collection and analysis, enabling us to gather thorough insights and address the diverse needs of users. Data will be collected through surveys, interviews, and observations. The surveys will deliver quantitative data regarding user experiences and requirements, targeting a sample size of around 50 students and 12 lecturers to ensure a representative understanding of the user base. Qualitative insights into specific challenges will be gained through interviews with frequent bus users and administrative staff. Additionally, observations of the current bus service will assist in identifying logistical problems. Quantitative data analysis will utilize Google Forms' graphical representation tools for summarization and interpretation, while qualitative data will be examined and categorized to identify themes and patterns. The development of the system will be conducted through an iterative approach that encompasses design, coding, testing, and deployment stages. This will utilize technologies such as GPS for real-time tracking and mobile platforms for online booking, employing JavaScript and React. User testing will be implemented in several phases, gathering feedback from students and lecturers who regularly utilize the bus service, which will inform adjustments based on usability studies and focus groups. The project will adhere to a structured timeline that includes phases for data collection, system development, user testing, and final deployment. Ethical considerations will be prioritized by safeguarding privacy and confidentiality, securing informed consent, and ensuring data anonymization. By adopting this thorough methodology, this strive to develop a Staff Bus Management System that is efficient, reliable, and user-friendly, catering to the needs of KDU students and lecturers.

9. Functional and Non-Functional Requirements

9.1. Functional Requirements

1. **Real-Time GPS Tracking**:
    i. The system should provide real-time location updates for all buses in the fleet.
    ii. Users should be able to view the current location of the bus on a map interface.
2. **Online Seat Booking**:
    i. Users should be able to reserve seats online through a web-based platform or mobile app.
    ii. The system should display available seats and allow users to book, cancel, or modify their reservations.
3. **Automated Notifications**:
    i. The system should send automated notifications to users about bus schedules, delays, cancellations, and other important updates via SMS or push notifications.
    ii. Users should be able to customize their notification preferences.
4. **Feedback and Reporting Mechanisms**:
    i. Users should be able to provide feedback and report issues through the platform.
    ii. The system should generate reports based on user feedback for continuous improvement.
5. **Emergency Alerts**:
    i. The system should include an emergency alert feature to notify users in case of any critical issues or emergencies during the bus journey.
6. **Ticket Rollover**:
    i. The system should allow users to rollover and reissue tickets that were not used for a particular week to a future date.

9.2. Non-Functional Requirements

1. **Scalability**:
    i. The system should be able to handle a growing number of users and increased data load without performance degradation.
    ii. It should support future expansion, including additional routes and more buses.
2. **Performance**:
    i. The system should provide quick response times for user queries and real-time updates.
    ii. It should handle concurrent user requests efficiently.
3. **Reliability**:
    i. The system should ensure high availability and minimal downtime.
    ii. It should include mechanisms for regular backups and disaster recovery.
4. **Security**:
    i. The system should implement strong security measures to protect user data and prevent unauthorized access.
    ii. It should comply with relevant data protection regulations and standards.
5. **Usability**:
    i. The system should have an intuitive and user-friendly interface.
    ii. It should be accessible to users with varying levels of technical proficiency.
6. **Maintainability**:
    i. The system should be designed for easy maintenance and updates.
    ii. It should include comprehensive documentation for developers and administrators.
7. **Privacy**:
    i. The system should ensure the privacy of user data, including personal and booking information.
    ii. It should include features for user consent and data anonymization where necessary.

10. Technology

The development of the Staff Bus Management System will leverage a range of technologies to ensure a robust and efficient solution.

10.1. Front-End Technologies:

i. **React:** For building the web application, React will provide a dynamic and responsive user interface, ensuring a seamless experience across various devices and screen sizes. Consumes the APIs provided by the Node.js backend to display data and interact with the server.

ii. **React Native:** This project are using react native cli, React Native CLI is the traditional approach for building React Native apps. For the mobile application, this will be used to develop a cross-platform app, allowing for a native-like experience on both iOS and Android devices with shared codebases. It provides maximum flexibility and control over this project setup and configuration.

10.2. Back-End Technologies:

i. **Node.js:** Utilized for the back end of the mobile application, Node.js will handle real-time data processing and server-side logic efficiently and it serves as the server-side environment to handle business logic, database interactions, and API endpoints.

ii. **Spring Boot:** For the web application back-end, Spring Boot will provide a robust framework for building scalable and secure server-side components.

10.3. Database:

i. **Firebase:** Chosen for its real-time data synchronization capabilities, Firebase will serve as the database solution, enabling seamless data management and real-time updates for both the web and mobile applications.

10.4. API and Dependencies:

i. **React native maps:**
React Native Maps is a library for React Native that allows developers to integrate maps into their mobile applications. It provides components and APIs for adding interactive maps, markers, and other map-related features. This library uses either Google Maps (on Android) or Apple Maps (on iOS) to render the maps, making it easy to display geographical data and locations within a React Native app.

ii. **React-Native Geolocation (Tracking):**
For tracking live location, the system need the react-native-geolocation-service and @react-native-community/geolocation libraries.

iii. **Directions API:**
Provides directions between locations. This is useful for building navigation features, such as generating step-by-step directions or calculating travel routes.

iv. **Geolocation API:**
Determines the geographical position (latitude and longitude) of a user's device. It's essential for tracking the user's real-time location, which is critical for apps needing location awareness.

v. **Places API:**
Provides detailed information about places, such as businesses, landmarks, or points of interest. It also supports searching for places within a specified area. The system uses this to search for locations near the user or provide suggestions based on keywords.

10.5. Authentication and Authorization:

i. Implement user authentication to secure access to the app and server.
ii. Use technologies like JSON Web Tokens (JWT) for token-based authentication.
iii. Ensure that only authorized users can access specific functionalities.
Example: login panel, registration panel, and admin panel

10.6. Version control:

i. **GitHub:** GitHub is a cloud-based platform. Group members can store, share, and work to get there with other code. This project hopes to use GitHub desktop.

11.Time Plan

	April				May				June				July				August				September				October				November			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Project Duration			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■				
Identifying the problem			■	■	■																											
Requirements Gathering						■	■	■																								
Documentation									■	■	■	■																				
Conducting software training											■	■	■	■																		
Designing Interfaces															■	■																
Designing UML diagrams																	■	■														
Back-end coding																	■	■	■	■	■	■	■	■	■							
Testing the system																										■	■					
Implementing the system																												■				
Creating final project report																												■				
Learning curve			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■				

12.Conclusion

The proposed Staff Bus Management System aims to tackle the current inefficiencies and challenges faced by the transportation service at Kotelawala Defence University (KDU) Southern Campus. By incorporating advanced features such as real-time GPS tracking, online seat booking, automated notifications, and feedback mechanisms, the system intends to enhance the commuting experience for both students and staff. The utilization of modern technologies including React, React Native, Node.js, Spring Boot, and Firebase will deliver a robust, scalable, and user-friendly solution. This project not only aims to bridge existing communication gaps and resolve logistical issues but also aligns with sustainability objectives by minimizing paper usage and enhancing operational efficiency. With a strong emphasis on user needs and technological advancement, the implementation of this system will greatly improve the reliability and effectiveness of the university's transport service, ultimately leading to a more efficient and satisfying commuting experience for the KDU community.

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